

LSTM Hyperparameter Tuning with Improved Particle Swarm Optimization for Stock Price Prediction

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1 TIMELINE

Table 1: Rough Timeline and Work Plan (Now – End of April)

Week	Dates	Planned work (deliverables)
W1	Mar 02 – Mar 08	Project proposal and planning
W2	Mar 09 – Mar 15	Spring Break
W3	Mar 16 – Mar 22	Implement LSTM model baseline
W4	Mar 23 – Mar 29	Implement PSO algorithm for hyperparameter search
W5	Mar 30 – Apr 05	Experimentation
W6	Apr 06 – Apr 12	Experiment analysis
W7	Apr 13 – Apr 19	Write final report
W8	Apr 20 – Apr 26	Final presentation
W9	Apr 27 – Apr 30	...

2 OVERVIEW

2.1 An Overview of the Proposed Study

This study investigates the efficacy of a Particle Swarm Optimization (PSO) algorithm for hyperparameter tuning of Long-Short-Term Memory (LSTM) networks in the context of stock price prediction. The primary objective is to explore the search capabilities of PSO on complex problems and to evaluate the

performance of the PSO-tuned LSTM against traditional tuning methods such as grid search. We expect that the PSO-tuned LSTM will outperform the baseline models in terms of predictive accuracy and trading performance. However, we also anticipate some potential challenges, such as premature convergence and overfitting to a local optima. If necessary, this will give us the opportunity to explore other PSO-adjacent approaches such as Firefly or Bat algorithms to further improve the optimization process.

2.2 Experimental Design

2.2.1 Data and Splits

The study utilizes a comprehensive 5-year historical dataset consisting of 1-minute interval data for 51 tickers (50 diverse equities representing various market sectors plus the SPY S&P500 ETF as a market benchmark). The target variable is the mid-price movement, calculated as the average of the best bid and best ask prices. The data will be split into a 3-year training set, a 1-year validation set (used for PSO fitness evaluation), and a 1-year out-of-sample testing set to verify the model’s ability to generalize across different market conditions.

2.2.2 Baselines

To evaluate the relative advantage of PSO-based tuning, the model will be compared against:

1. A **standard LSTM** tuned via traditional gradient-descent optimization (Adam/RMSprop) and manual search.
2. An **XGBoost (Extreme Gradient Boosting)** regressor to provide a non-deep learning machine learning baseline.
3. A **Persistence Model (Naive Forecast)**, where the predicted price is simply the current price, to establish a minimum threshold for predictive value.

2.2.3 PSO Search Space

The swarm will optimize a 5-dimensional space representing the following LSTM levers:

1. **Hidden Layers:** $\{1, 2, 3, 4\}$
The swarm evaluates if added depth improves directional accuracy or merely memorizes noise. It pulls the population toward the simplest robust solution.
2. **Neurons per Layer:** $[32, 512]$
Finds the "capacity" sweet spot to capture complex signals across 51 diverse tickers without redundant co-adaptation.

3. **Learning Rate:** $[10^{-5}, 10^{-1}]$
Simultaneously tests multiple step-sizes to avoid overshooting in the high-volatility HFT loss landscape.
4. **Dropout Rate:** $[0.0, 0.5]$
Optimizes for generalization, prioritizing particles that maintain high performance on unseen validation data.
5. **Lookback Window:** $\{10, 30, 60, 120\}$ **min**
Determines the optimal temporal context, discarding "stale" data that no longer impacts the next-minute price action.

2.2.4 Metrics

The following metrics will be used to guide the PSO fitness function and evaluate the final "Global Best" model:

Type	Metric	Symbol / Definition	Obj.	Rationale
Trading	Sharpe Ratio	$SR = \frac{E[R_p - R_f]}{\sigma(R_p - R_f)}$	Max	Primary measure of risk-adjusted return.
Trading	Max Drawdown	$MDD = \frac{\max(Peak - Trough)}{Peak}$	Min	Measures the worst peak-to-trough decline (risk).
Trading	Profit Factor	$PF = \frac{\text{Total Gross Profit}}{\text{Total Gross Loss}}$	Max	Measures the quality of the trading system's returns.
Trading	CAGR	Geometric mean of annual returns.	Max	Reflects the average annual growth of the investment.
Statistical	RMSE	$\sqrt{E[(\hat{y} - y)^2]}$	Min	Measures the magnitude of the mid-price prediction error.
Statistical	Directional Acc.	Correct sign predictions / Total	Max	Measures the ability to predict market momentum.
Statistical	F1 Score	$2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$	Max	Robust for imbalanced ternary (up/down/flat) classes.
Statistical	AUC (Micro)	Area Under the ROC Curve	Max	Measures discriminative power independent of threshold.

3 MOTIVATION

3.1 Motivation and Relevance to Course

The global financial and economic activities can be negatively impacted by the volatility of the stock market [Bloom(2009)]. Stock price forecasting is usually employed as a tool for setting investment strategies. One of the most popular machine learning algorithms for stock price forecasting is the long-short-term memory (LSTM) [Lanbouri and Achchab(2020)]. LSTM forecasters, however, are sensitive to hyperparameters, and manual tuning is often slow and inefficient. Nature-inspired collective behavior motivate metaheuristics that efficiently search complex, noisy and nonlinear landscapes that typically characterize stock price movements. In this project, we use improved particle swarm optimization (PSO) as a bio-inspired optimizer to automate hyperparameter selection of LSTM for stock price prediction [Deng and Peng(2025)]. The PSO algorithm, however, could run into premature convergence and end up at the local optimum [Ji et al.(2021)]. Consequently, we further plan to explore some approaches that could be used to improve PSO.

3.2 Research Questions

At the end of this project, we hope to be able to answer the following questions:

- Does improved PSO produce lower forecast error than baselines under fixed evaluation budget?
- Which hyperparameters affect accuracy and stability the most?

References

- [Bloom(2009)] Nicholas Bloom. 2009. The Impact of Uncertainty Shocks. *Econometrica* (2009). <https://doi.org/10.3982/ECTA6248>
- [Deng and Peng(2025)] Chao Deng and Jiao Peng. 2025. Application of LSTM Model based on Particle Swarm Optimization Algorithm in Stock Market Trend Prediction. In *Proceedings of the 2nd Guangdong-Hong Kong-Macao Greater Bay Area International Conference on Digital Economy and Artificial Intelligence (DEAI '25)*. Association for Computing Machinery, Dongguan, China. doi:10.1145/3745238.3745261
- [Ji et al.(2021)] Yi Ji, Alan Wee-Chung Liew, and Lixia Yang. 2021. A Novel Improved Particle Swarm Optimization With Long-Short Term Memory Hybrid Model for Stock Indices Forecast. *IEEE Access* 9 (2021), 23660–23671. doi:10.1109/ACCESS.2021.3056713
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